



Detecting the impact of bank and channel modification on invertebrate communities in Mediterranean temporary streams (Sardinia, SW Italy)



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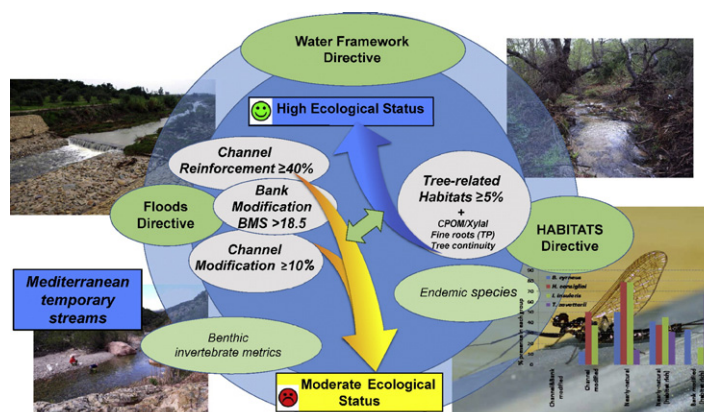
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HIGHLIGHTS

- Bank and channel modifications affect benthic community and endemic species.
- Tree-related habitats mitigate the impact of stream degradation on invertebrates.
- Benthic metrics for ecological status detect the effects of morphological alteration.
- Habitat and modification thresholds are provided for management measures.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 30 March 2016

Received in revised form 20 May 2016

Accepted 21 May 2016

Available online 28 May 2016

Editor: D. Barcelo

Keywords:

Benthic taxa
Tree-related habitat
Ecological status
Endemism
WFD
MRT

ABSTRACT

We hypothesized that reach-scale, bank and channel modification would impact benthic communities in temporary rivers of Sardinia, when pollution and water abstraction are not relevant. A range of variables were considered, which include both artificial structures/alterations and natural features observed in a stream reach. Multivariate regression trees (MRT) were used to assess the effects of the explanatory variables on invertebrate assemblages and five groups, characterized by different habitat modification and/or features, were recognized. Four node variables determined the splits in the MRT analysis: channel reinforcement, tree-related bank and channel habitats, channel modification and bank modification. Continuity of trees in the river corridor diverged among MRT groups and significant differences among groups include presence of alders, extent of channel shading and substrate diversity. Also, the percentage of in-stream organic substrates, in particular CPOM/Xylal, showed highly significant differences among groups. For practical applications, thresholds for the extent of channel reinforcement (40%) and modification (10%) and for bank alteration ($\approx 30\%$) were provided, that can be used to guide the implementation of restoration measures. In moderately altered river reaches, a significant extent of tree-related habitats ($\approx 5\%$) can noticeably mitigate the effects of morphological alteration on aquatic invertebrates. The outcomes highlight the importance of riparian zone management as an opportune, achievable prospect in the restoration of Mediterranean temporary streams. The impact of bank and channel modification on

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ecological status (sensu WFD) was investigated and the tested benthic metrics, especially those based on abundance data, showed legible differences among MRT groups. Finally, bank and channel modification appears to be a potential threat for the conservation of a few Sardo-Corsican endemic species. The introduction of management criteria that openly use information on endemic species seems crucial to reduce the risk of improving overall ecological status while depleting biodiversity.

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1. Introduction

Bank alteration and channelization are widespread in the Mediterranean context and notably affect the majority of river systems, including least-disturbed streams (Feio et al., 2014). As well, infrastructure works and road crossings, such as bridges and culverts, often combined to grade control structures, can alter habitats and disrupt river and stream continuity (Litvan et al., 2008; Manariotis and Yannopoulos, 2014). Furthermore, in agricultural and urban landscapes, riparian vegetation is often confined to river banks and banks are resectioned or reinforced to expand the cultivated area and/or to avoid flooding (Feio et al., 2014). Armouring of streambanks, that is a common management answer to perceived threats to adjacent infrastructure or fields from flooding or erosion, is a critical example of ubiquitous cause of ecosystem impairment (e.g. Stein et al., 2011, 2013).

The relevance of these issues is real and well recognized by Policy makers. An example at the European level, where two distinct but inter-connected Directives were promulgated, which directly or indirectly deal with them. These are the Floods Directive (FD: EC, 2007) and the Water Framework Directive (WFD: EC, 2000). The FD provides guidance on sustainable flood protection management, thus directly dealing with bank armouring and stabilization. The WFD aims at achieving good ecological and chemical status of all surface water bodies and requires hydromorphological data to assist the interpretation of biological response. These Directives must, in turn, interrelate with the HABITATS Directive (HD: EC, 1992), whose aim is to promote the maintenance of biodiversity, through the protection of natural habitats and species. The harmonized application of such directives is essential, especially where very high values of endemism richness are observed (Kier et al., 2009), such as the Sardo-Corsican system, that exhibits the highest proportion of endemic species in Italy (Stoch, 2000).

Nonetheless, reach-scale bank armouring has received less attention than other hydromorphological modification and the mechanisms and magnitude of its effects on river communities have been studied briefly and are poorly known (Stein et al., 2013). In this regard, an improved understanding of the ecological impact of bank armouring structures would enhance our ability to support watershed managers in the protection or restoration of aquatic resources (Stein et al., 2013). Side effects of bank armouring and stabilization are often a limited riparian function and decreased heterogeneity of riparian and in-channel natural habitats (Florsheim et al., 2008). Finally, richness and attributes of such habitats largely control animal species variety and aquatic communities (e.g. Greenwood et al., 2012; Tews et al., 2004).

A critical aspect of studies aimed to analyse the effects of hydromorphological alterations on river biota is the frequent occurrence of multiple stressors (Friberg et al., 2009; Friberg, 2014). These authors argued it is presently not possible to quantify and rank the importance of hydromorphology to that of other stressors, and concluded by hoping for future studies that address this issue. Indeed, excluding cases of multiple stressors, most studies devoted to distinctively assess the effects of habitat degradation on aquatic invertebrates in rivers often focus on somewhat obvious scenarios, where hydromorphological degradation is heavy and its consequences are therefore easily foreseeable (e.g. Mueller et al., 2011; Wyżga et al., 2014). While this kind of studies provides useful evidence on the effects of localized and specific river modification, it is still necessary for investigations to be conducted specifically for situations where alteration is scattered (e.g. resectioning, armouring) and subtler deviation from naturalness is observed.

Mediterranean rivers are often subjected not only to nested anthropogenic stresses but also to naturally demanding conditions. Actually, in Mediterranean landscapes, the effects of infrequent torrential rain events on erosion, flooding and stream channel changes (Poesen and Hooke, 1997) can be conceived as dominant factors in shaping ecological communities in rivers. The same can be said for the occurrence of dry periods, whose duration and extent are often variable over time (e.g. De Girolamo et al., 2015). These overriding factors often overlay point source pollution and water abstraction, and call into question the possibility that any reach-scale effect of bank and channel modification on benthic communities can be discovered.

Focusing on habitat features that can be directly subjected to management practices, we specifically hypothesized that at reach-scale, bank and channel modification would actually impact benthic communities in rivers of Sardinia when pollution and water abstraction are not observed. The approach we adopted aims at answering the following questions: a) “Do benthic communities reflect bank and channel modification in Mediterranean temporary rivers?”, b) “Which habitats are more relevant in shaping benthic communities?”, c) “Does bank and channel modification affect ecological status for aquatic invertebrates?”, d) “Do endemic species mirror habitat features and river alteration?” and e) “Can we improve the effectiveness of stream corridor management practices for aquatic invertebrates?”.

2. Materials and methods

2.1. Study area

The study was conducted in Sardinia, South Western Italy. The second largest island in the Mediterranean Sea, Sardinia presents a Mediterranean climate characterized by mild, relatively rainy winters and dry, hot summers. Sardinian rivers often present typical Mediterranean character, being commonly characterized by the irregularity of flow and strong seasonal hydrological fluctuations (De Waele et al., 2010). Prevailing impacts on rivers related to alteration of water quality commonly derive from untreated civil sewage, agricultural practices and farming (RAS, 2005). Local management practices and morphological alteration, which are often due to road crossings or flood defence and water abstraction needs, include, among others, bank resectioning, bank reinforcement and channel stabilization.

Investigated river reaches are spread over a wide area, mainly in the eastern side of the island, encompassing the whole north – south axis. The list of investigated sites is reported in Table S1 (SM - Supplementary material). Most sites are located on temporary water bodies according to the official regional typology (RAS, 2009). The few water bodies classified as perennial retain however a strong Mediterranean character and show major flow reduction during the spring-summer season experiencing droughts in extremely dry years. Catchment size is comparable among study reaches, that belong to small streams (Table S4, SM).

2.2. Site selection

Sites have been selected according to hydromorphological and habitat features in order to cover a wide alteration gradient. River reaches affected by relevant water pollution were removed from dataset. Particularly, the samples used in the present analysis belong to water bodies classified from High to Moderate according to LIMeco descriptor (DM 260/2010, MATTM, 2010), formally used for a chemistry-based

classification of water bodies in Italy (see Table S4, SM). In addition, the selected samples concurrently match at least three of the following conditions: orthophosphate $<300 \mu\text{g l}^{-1}$; electrical conductivity $<800 \mu\text{S cm}^{-1}$; chloride $<150 \text{mg l}^{-1}$ and ammonia $<0.05 \text{mg N l}^{-1}$. Lastly, sites downstream from relevant water intakes or sampled shortly after drought events were excluded. There were 62 sites upon completion of selection and these were included for data analysis. Biological and environmental data were simultaneously collected in February–August 2004 (26 sites), May 2011 (30 sites) and March 2013 (6 sites). To the author's knowledge, none of the sampling periods occurred after recent flooding events.

2.3. Habitat and stream reach characterization

Habitat features and river modifications were collected at reach scale by means of the CARAVAGGIO method (Buffagni and Kemp, 2002; Buffagni et al., 2013), developed on the basis of the River Habitat Survey protocol (RHS, Raven et al., 1997). The method is based on a comprehensive survey, with data collected for 500 m channel lengths. Survey is undertaken on channel, banks and peri-fluvial areas following a two-section protocol. In the first part, bank and channel features are recorded every 50 m on 10 $\approx$ equally spaced spot-checks perpendicular to river flow; in the second section (sweep-up), features are recorded along the whole 500 m stretch. Qualitative, quantitative and categorical information is collected, depending on the kind of feature (e.g. Buffagni et al., 2010; Raven et al., 1997). In addition to single habitat features, based on the collected information, the following habitat descriptors were calculated and included in the data analysis: HMS (Habitat Modification Score – presence and quantity of artificial features along river channel and banks) and HQA (Habitat Quality Assessment score – measure of habitat richness and diversification) (Raven et al., 1998), LUir (Land Use Index (reach scale) – quantification of land use modification at the reach scale) (Erba et al., 2015), IQH (average of the previous three) (Buffagni et al., 2013), LRD (Lentic-lotic River Descriptor) (Buffagni et al., 2010). CARAVAGGIO survey was undertaken simultaneously to benthic invertebrate collection. The 500 m reach extends 400 m upstream and 100 m downstream the macroinvertebrate sampling site. Map-based geographical data were collected for each site and include distance from source and slope of the thalweg. Water discharge was gaged during sampling with the velocity-area method (Buchanan and Somers, 1969) with stream water velocity measured using a current meter, at a number (≥ 2) of representative cross-sections of the stream.

2.4. Biological data

Macroinvertebrates were collected according to a sampling procedure based on a multi-habitat scheme where organic and mineral habitats, respectively classified according to the biotic type and to substrate size, are sampled in proportion to their occurrence within a 25–50 m reach. Sampling units are identified at a riffle-pool sequence within the reach (Buffagni et al., 2001) and 10 sampling units are proportionally collected in pool and 10 in riffle (i.e. same sampling effort in pool and riffle), both including marginal habitats. Sampling units are quantitatively collected with a Surber net (area 0.05m^2 ; mesh size 0.5mm). According to the national standards for WFD ecological status classification (Buffagni and Erba, 2007; MATTM, 2010), centred on the full representativeness of invertebrates collection from the pool mesohabitat, only samples from pool areas were considered for core analysis in the present study. In fact, for Mediterranean rivers, the National Italian assessment system (Buffagni et al., 2008) always requires a sample to be collected from the pool mesohabitat for ecological classification, while the sample from riffles (not permanently found in such river types) is collected only at certain circumstances. Therefore, to guarantee a wide applicability of the results from the present paper, focus was placed on pool communities. The choice of concentrating on

pools is also based on Buffagni et al. (2004) who concluded that the benthic metrics used for classification show a better performance in pools than in riffles in Italian Mediterranean streams. Additionally, Monaghan and Soares (2010) proved that standing-water samples reflected more strongly habitat characteristics at the reach scale than running water samples, corroborating the selection of pool mesohabitat for sampling.

In order to provide results appropriate at a wider geographical scale and considering the high number of collected families, data analyses were performed on family level taxa lists. Furthermore, prior to multivariate analysis, taxa lists were reduced by excluding taxa present in $<5\%$ and $>75\%$ of the samples. This adjustment was deemed as convenient in order to exclude rare and ubiquitous (and often dominant) families. For multivariate analysis, after a logarithmic ($\log_{10}$) transformation, invertebrate abundance data from each sample were transformed into a vector of length 1 using the following equation (Legendre & Legendre, 2012, eq. 7.67):

$$y'_{ij} = \frac{y_{ij}}{\sqrt{\sum_{j=1}^p y_{ij}^2}}, \text{ where } y_{ij} \text{ is the abundance of family } j \text{ in sample } i,$$

across the p families. For a selection of endemic taxa (see Section 2.5.5), when testing differences in distribution among multivariate MRT groups, species level was maintained.

For each site, the following biological metrics were calculated at the family level on a complete taxa list for the pool mesohabitat: ASPT (Average Score per Taxon; Armitage et al., 1983); N_families (Total Number of Families; e.g. Thorne and Williams, 1997); EPT (number of families of Ephemeroptera, Plecoptera and Trichoptera; Barbour et al., 1999); Sel_EPTD ($\log_{10}$ [sum of abundance of Heptageniidae, Ephemeridae, Leptophlebiidae, Brachycentridae, Goeridae, Polycentropodidae, Limnephilidae, Odontoceridae, Dolichopodidae, Stratiomyidae, Dixidae, Empididae, Athericidae & Nemouridae + 1]; Buffagni et al., 2007); Shannon (Shannon-Wiener Diversity index; Shannon and Weaver, 1949); 1-GOLD (1 - relative abundance of Gastropoda, Oligochaeta and Diptera; Pinto et al., 2004). These six metrics were combined (weighted average) in the STAR_ICM index (STAR Intercalibration Common Metric index, Buffagni et al., 2007), designed to assess general degradation (Erba et al., 2009) and officially adopted for ecological status classification according to macroinvertebrate fauna in Italy (MATTM, 2010). MacOper.ICM software (ver. 1.0.5, Buffagni and Belfiore, 2013) was used for metrics calculation.

2.5. Data analysis

2.5.1. Explanatory, test and biological variables

The analyses performed deal with four theoretical categories of variables: 1) Explanatory variables used in multivariate, clustering analysis (MRT, see Section 2.5.2) to constrain (i.e. where to divide) the groups found based on benthic taxa. They include i) artificial structures/alterations observed in a stream reach and ii) natural features that can be, ideally, fully or partly controlled for management/restoration purposes. These variables (Table 1) are supposed to be appropriate for establishing the link between local river management practices and biological response. 2) Test variables i.e. environmental features useful to clarify meaning of multivariate groups, indirect links between explanatory variables and biocoenosis or relevance of background variables. This set of variables (Table 2 and Table S3, SM) mainly includes habitat and land-use features. Also, geographical, stream type, water quality variables and CARAVAGGIO habitat descriptors are comprised. 3) Biological metrics for benthic invertebrates (Section 2.4 and Table 3) investigated to examine the potential association of MRT groups and corresponding habitat alteration/feature with ecological status of the water body, sensu WFD. 4) A range of endemic taxa to test possible concern of MRT groups and explanatory variables for biodiversity issues and conservation. The range of analytical procedures and logical steps adopted in the paper are shown in Fig. S2 (SM) in direct connection to the aims specified in this paper.

Table 1

Potential explanatory variables included in MRT analysis. Some variables combine elements already present in others. All features are recorded along a 500 m long river reach.

Category	Riverine feature, description and specification		Calculation	
Artificial structures/river modification	Channel resectioning	Channel cross section modification from its natural form to a standard form, typically associated with straightening and channelization; excludes channel reinforcement	%	Occurrence at 10 $\approx$ equidistant transects
	Channel reinforcement	Channel stabilization through armouring that may add to channel reprofiling		
	Channel modification	Includes: channel resectioning and reinforcement, culverts, fords, weirs and sluices		
	Bank resectioning	Bank reprofiling without armouring	%	Occurrence on both banks at 10 $\approx$ equidistant transects
	Bank protection	Various types of bank armouring. Includes: whole bank and toe-only reinforcement; excludes top-only reinforcement		
	Bank modification - BMS	Includes: features of both Bank resectioning and Bank protection, bridges with bank side abutments and culverts	Score	Calculated as: $BMS = (\% \text{ Bank protection} + \% \text{ bank resectioning} * 0.5 + \text{number of Bridges} * 5 + \text{number of Culverts} * 5) / 1.65$. Values range from 0 to 100+
	Bridges	Includes: bridges with in-channel supports and/or bank-side abutments	Count	Number of structures in the whole reach
Natural features	Culverts	Closed conduits generally used to convey water for stream crossing or run-off management		
	Tree cover continuity	Continuity of woody vegetation along banks (mainly trees, ≥ 2 m high)	Score	Arithmetic mean of extent on left and right bank, after numeric conversion (none: 0, Isolated/scattered: 1, regularly spaced-single: 2, occasional clumps: 3, semi-continuous: 4, continuous: 5)
	Tree-related bank habitats	Includes: overhanging boughs, exposed bankside roots and fallen/leaning trees on the lower bank	%	Average percentage of observed features included in the variable. The estimated extension of each feature is reported as follows: present: 10%, extensive: 40%, whole stretch: 90%
	Tree-related channel habitats	Includes: large woody debris and fallen trees (within the channel)		
Bank erosion	Tree-related bank and channel habitats	Includes: all features of both tree-related bank and channel habitats		
	Erosional features along banks	Various types of erosion-related bank features. Includes: eroding bank (whole), eroding bank (toe only), bank toe and concave bars	%	Occurrence at ≈ 10 equidistant transects on both banks
	Absence of bank erosion	Portion of banks not showing erosional features	%	Occurrence of transects (out of 10 on both banks) lacking bank erosion

2.5.2. Grouping of river reaches according to benthic community and habitat variables

Multivariate regression trees (MRT: De'ath, 2002) were used to assess the hierarchical effects of the explanatory variables (Table 1) on invertebrate composition assemblages (i.e. the response data set). MRT clustering creates a hierarchical tree with successive dichotomies in function of the selected variables, with related invertebrate assemblages for each tree partition, and makes no assumptions about the form of relationships between taxa and their environment. The relative error (RE) is the variation among samples not described by the tree and represents the fit of the tree. Among the arguments, due to the small number of observations, we used as many cross validation groups as rows in the data set (62), and 100 iterations. The minimum number of cases for each terminal node was set at four samples. To find indicator taxa and define invertebrate assemblages characterizing groups of sites (i.e. each branch of the MRT tree), the Dufrêne–Legendre index (IndVal: Dufrêne and Legendre, 1997) was used, within the Indicator Species Analysis (ISA). The IndVal index is a combination of two sub-indices, accounting for taxon specificity (relative abundance) and fidelity (frequency of occurrence), and is based only on within-taxon abundance and occurrence comparisons (Legendre, 2013); its value is not affected by the abundances of other taxa. The index is maximum (i.e. its value is 1) when the individuals of a given taxon are observed at all sites belonging to a single cluster. Statistical significance (here fixed at 0.05) was assessed by Monte Carlo permutations procedure (9999 runs). In this study, MRT and IndVal analyses were carried out by using, respectively, the 'mvpart' (Therneau et al., 2014) and the 'MVPARTwrap' (Ouellette and Legendre, 2014) packages. Analyses were performed in R3.1.1 (R Core Team, 2014).

2.5.3. Comparing test variables among MRT groups

The nonparametric BDM test (Brunner et al., 2012) was used to test rank differences in habitat and environmental variables among MRT groups (R package 'asbio', Aho, 2014, 2015). This test can be regarded as a robust extension of the classical ANOVA to heteroscedastic designs with unequal cell frequencies, and also applies in nonparametric designs, because rank statistics are heteroscedastic in general (Akritas, 1990). Due to the relatively high number of variables tested, False Discovery Rate (FDR) was controlled by adjusting the p-level values obtained for the range of variables with the Benjamini and Yekutieli (2001) correction (function 'p.adjust', R Core Team, 2014). For significant variables, post-hoc comparison was performed with Dunn's (1964), with the Benjamini and Hochberg (1995) correction for FDR (R package 'dunnTest', Dinno, 2015).

Spearman's rank coefficient was used for correlation among variables (Table S7, SM).

2.5.4. Comparing benthic metrics for MRT groups against reference conditions

For biological metrics used in ecological status assessment, rank-based multiple comparison and compatible simultaneous confidence intervals (SCI) for transitive relative effects in unbalanced one-way designs were performed (Konietschke et al., 2012, 2015). Dunnett's test (Dunnett, 1955) for many-to-one comparisons (confidence level: 95%; Fisher method) was applied ('mcp' function in R package 'nparcomp', Konietschke et al., 2012, 2015), which does not assume homogeneous variances. MRT groups were compared to the pool of reference sites ($n = 17$) sensu WFD formally accepted for Sardinia, and comprised in the study dataset.

Table 2
Results of nonparametric ANOVA (BDM test) among MRT groups for selected explanatory and test variables. Significance after FDR correction: ***p < 0.001; **p < 0.005; *p < 0.05; (*) p < 0.1. Outcomes of post-hoc comparisons (Dunn's test) are also reported, with an indication of the group(s) showing higher (+) and lower (–) values for the variable. ANOVA results for the full set of tested variables are provided in Table S3 of the Supplementary material.

Variable category	Variable	BDM test F	p-Level _{fdr}	Post-hoc significance
Habitat	Alders	9.5	0.014	*
	Channel shading of wetted channel	17.3	0.000	***
	Tree cover continuity	21.8	0.001	**
	Substrate %	7.1	0.102	^a
	Sum of gravel, pebbles and cobbles (0.2–25.6 cm)	22.5	0.000	***
	Sum of CPOM and Xylal	6.9	0.097	(*)
	Sum of Submerged macrophytes, living parts of terrestrial plants and mosses	9.6	0.060	(*)
	Diversity (Shannon-Wiener index)	7.6	0.060	(*)
	Organic substrates (include the two categories above and algae and emergent macrophytes)	48.0	0.000	***
	All substrates (mineral and organic)	11.0	0.040	*
Natural land-use	Habitat descriptors (CARAVAGGIO and RHS)	46.2	0.000	***
	HMS - Habitat Modification Score	23.4	0.000	***
	HQA - Habitat Quality Assessment score	9.9	0.003	**
	IQH - Index of Habitat Quality	9.0	0.013	*
Other	LUIr - Land Use Index reach	11.0	0.001	**
	Bankface land-use %	11.0	0.001	**
	Land-use in adjacent lands %	9.9	0.003	**
Other	Slope of the thalweg	11.0	0.001	**
	Sparse vegetation (tall herbs, rank vegetation and grassland)	9.0	0.013	*

^a Highly significant before FDR correction.

2.5.5. Endemic species testing

Eight endemic species were found in a sufficient number of sites (i.e. ≥ 10 , $\approx 15\%$) and tested for differences among MRT groups (see Table 4). Taxa are reported hereafter. Ephemeroptera: *Baetis cyrneus* Thomas and Gazagnes 1984, *Baetis* cf. *albinatii* Sartori and Thomas 1989, *Baetis* cf. *ingridae* Thomas and Soldan 1987, *Electrogena zebrata* (Hagen, 1864) and *Habrophlebia consilgloi* Biancheri 1959. Plecoptera: *Tyrrhenoleuctra zavattarii* (Consiglio, 1956) and *Isoperla insularis* (Morton, 1930). Trichoptera: *Agapetus cynensis* Mosely, 1930. For *Baetis* species of the *muticus* and *rhodani* groups (i.e. *B.* cf. *albinatii* and *B.* cf. *ingridae*, respectively) more than one species can be present in the studied collections (Bisconti et al., Unpublished data). Selected taxa are confined to Sardinia and Corsica (Buffagni et al., 2009; de Jong, 2013;

Fochetti et al., 1998, 2009; Graf et al., 2008), in some cases to a few adjacent areas as well (Buffagni et al., 2003; Cianficconi and Moretti, 1990; Zrelli et al., 2011). The eight taxa (Table 4) were tested by means of a G^2 log-likelihood-ratio statistic (Dunning, 1993; Sokal and Rohlf, 1981): $G^2 = O_j \cdot \ln(O_j/E_j)$, as a measure of strength of association of taxa to MRT groups. The observed taxon frequencies (O_j) were compared with sampling frequencies in each group (E_j). Because G^2 can be unreliable if expected frequencies of <5 are involved (Moore, 2004), when the presence of a species in a group was <5 , that group was not incorporated for testing. Because invertebrates larvae and nymphs may potentially move from riffles to pools and vice versa to colonize suitable micro-habitat, potential species disappearance has to be evaluated considering the largest possible range of habitats occurring at a site. Therefore, strength

Table 3
Mean values and results of comparison (Dunn's test) between MRT groups and reference sites for ecological status benthic metrics: *p < 0.05; (*) p < 0.1. Percentage of sites classified in high, good and moderate ecological status is reported in the last columns.

ASPT			N_families		EPT		Sel_EPTD		Shannon		1-GOLD		STAR_ICMi		Ecological status (%)			
MRT group/REF sites	n	Mean	p-Value	Mean	p-Value	Mean	p-Value	Mean	p-Value	Mean	p-Value	Mean	p-Value	Mean	p-Value	High	Good	Mod.
cb_mod	5	5.42	0.182	24.8	0.265	6.4	0.023 *	1.09	0.017 *	1.23	0.247	0.23	0.006 *	0.71	0.012 *	0	40	60
ch_mod	8	5.53	0.133	24.8	0.760	7.3	0.119	1.00	0.009 *	1.53	0.044 *	0.54	0.896	0.75	0.098 (*)	12	25	63
cb_nat	8	5.87	0.994	26.8	0.658	8.2	0.810	2.27	1.000	1.60	0.186	0.72	0.777	0.96	0.983	50	38	12
CB_NAT	18	5.92	1.000	30.4	0.753	9.6	0.994	1.83	0.469	2.10	0.982	0.70	0.918	1.01	0.972	67	33	0
BA_MOD	6	5.70	0.817	23.5	0.637	8.0	0.628	0.97	0.054 (*)	1.93	0.873	0.54	0.376	0.80	0.319	33.3	33.3	33.3
Reference sites	17	5.93	–	28.3	–	9.9	–	2.04	–	2.02	–	0.69	–	0.99	–	59	41	0

Table 4

Strength of association of endemic taxa to MRT groups (please refer to caption of Fig. 1 for codes). For each taxon, the following parameters are reported: number of samples where the taxon was collected (n), value of G^2 statistic, p level, MRT group associated to significant and positive residuals and MRT group(s) where the taxon was not observed. In a few cases (p value in brackets), analysis of residuals led to significant differences while G^2 test was not significant.

	<i>B. cf. albinatii</i>	<i>B. cyrneus</i>	<i>B. cf. ingradae</i>	<i>E. zebrata</i>	<i>H. consiglioi</i>	<i>I. insularis</i>	<i>T. zavattarii</i>	<i>A. cynensis</i>
n_(tot = 62)	52	22	58	49	26	28	15	46
G^2	3.72	6.15	0.13	0.75	13.01	8.88	10.48	0.48
p level	ns	ns	ns	ns	<0.05	(<0.1)	<0.05	ns
Residuals (+)					cb_nat p < 0.05	(cb_nat) (<0.1)	BA_MOD p < 0.05	
Absence		cb_mod			cb_mod, BA_MOD	cb_mod	cb_mod, ch_mod	

of association of endemic taxa to MRT groups was tested considering altogether the sampling units collected at each site (both pool and riffle mesohabitats) in order to account for all major habitats found in the reach (e.g. riffles, pools, runs, margins).

3. Results

3.1. Overall occurrence of environmental features and macrobenthic taxa in the investigated sites

The presence and extent of artificial structures and natural features (Table 1), considered as potential explanatory variables, were very diversified among the investigated river reaches. Channel and bank modifications were observed respectively in 40% and in 65% of sites. When present, their percentage along the reach varied from 2 to 100%. Bank resectioning was recorded in about one half of the sites (52%), ranging from 5 to 100% of the reach. At least one bridge was present in 37% of the sites while culverts were recorded in 16% of sites. As far as natural features are concerned, tree continuity along banks was observed in all the possible extent categories (i.e. isolated/scattered, regularly spaced/single, occasional clumps, semi-continuous and continuous). Several sites (55%) showed semi-continuous or continuous tree cover. Shading of wetted channel was also present with a high degree of variability, from absent to extended to the whole reach, as was the presence of tree-related bank and channel habitats. All features included in tree-

related bank and channel habitats (Table 1) were recorded at least once in the investigated sites. The most frequently recorded habitat was overhanging boughs, observed in 66% of the sites; the least frequent was fallen/leaning trees on the lower bank, recorded in 21% of the sites. Descriptive values of variables for MRT groups are reported in Table S4, SM.

A total of 106 macroinvertebrate families were present in the whole set of river sites in the region, of which 95 were collected in the 62 investigated sites. There were 63 families which were eventually included in the analyses after having excluded those present in <5% (n = 22) and >75% (n = 10) of the samples. The complete list of included and excluded families is presented in Table S5, SM.

3.2. Response of benthic communities to bank and channel modification and the importance of habitat features

3.2.1. MRT groups, node-relevant environmental variables and invertebrate assemblages

The multivariate regression tree analysis (MRT, Fig. 1) clustered the investigated river and reaches into five groups, constraining taxa to four environmental and habitat variables (node variables). Globally, the MRT model explained 21% of the overall biological variance (RE = 0.79), with a standard error of 0.085. In general terms, the lower RE (and cross validated RE) values, the better the tree for the data set under investigation. In this study, the selected tree is the best predictive tree (five-

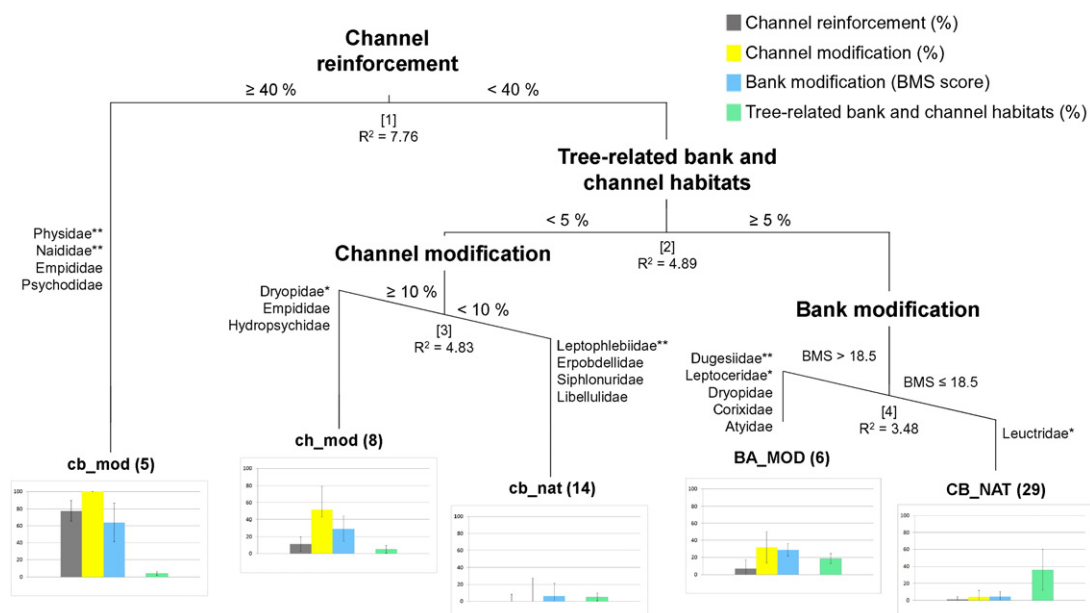


Fig. 1. Multivariate regression tree (MRT) obtained for Sardinian river sites. The resulting five impact groups are identified by codes summarizing prevalent features/alteration: cb_mod (altered channel and banks), ch_mod (altered channel), cb_nat (natural channel and banks), BA_MOD (altered banks) and CB_NAT (natural channel and banks). Codes in capital letters identify groups showing higher percentage of tree-related habitats. Groups in the foreground include least impaired sites. For each node variable, discriminant values are provided for the relevant dichotomy. In brackets, the number of sites. Predicted discriminant and indicator taxa are reported: * = discriminant taxon identified by MRT analysis; ** = taxon identified by both MRT and ISA analysis; no symbol = ISA only taxon. Histograms at the bottom of the tree show the mean values of the four node-relevant environmental variables in MRT groups (bars represent standard deviation). BMS: Bank modification score.

group solution) with lowest RE (and CVRE), besides providing a scenario with enough detail for interpreting benthic groups. A five letter code was assigned to each MRT group, which summarizes its main features/alteration (please refer to Fig. 1 caption). Codes in capital letters identify the two groups showing a higher percentage of tree-related habitats. Mean values of the four node-relevant environmental variables in MRT groups are reported in the histograms at the bottom of the tree in Fig. 1. The reported threshold values for node variables at each split are calculated as mean value between maximum and minimum observed in sites groups of the two branches. As they closely depend on the data set, such thresholds have to be considered as approximate (see Table S4, SM for observed ranges). Indicator species analysis (ISA), coupled with MRT analysis (Table S6, SM), selected a set of 25 characteristic taxa as best indicators (indicator values threshold: >25%), some of which showed up in Fig. 1.

The percentage of channel reinforcement defines the first MRT split (R^2 7.76%), parting the sites showing $\geq 40\%$ of alteration (group *cb_mod*) from the others. Physidae (11.5%) and Naididae (71.6%) distinguish the left branch. Sites with a higher percentage of channel reinforcement are characterized by Physidae, Naididae, Empididae and Psychodidae. The percentage of tree-related bank and channel habitats is the key environmental variable defining the split at node two (R^2 4.89%) in which Leptophlebiidae (30.9%), on the left, and Leuctridae (13.4%), on the right, gave the best contribution. IndVal analysis identified many taxa as a possible indicator (Table S6, SM). At node three (R^2 4.83%), the percentage of channel modification emerged as a relevant explanatory variable. Sites with higher channel modification ($\geq 10\%$, *ch_mod*) are defined by two indicator taxa (i.e. Hydropsychidae and Empididae) and by Dryopidae (larvae) as discriminant taxon (10.7%). More taxa characterize sites with channel modification $< 10\%$ (group *cb_nat*) with Leptophlebiidae as the only discriminant taxon (61.2%). Bank modification (BMS), mainly in terms of resectioning (Table S4, SM), discriminates invertebrate assemblages at node four (R^2 3.48%), with Corixidae, Atyidae, Dugesidae and Dryopidae (larvae) representative for sites affected by higher percentage of bank modification (BA_MOD) and Dugesidae (32.9%) and Leptoceridae (20.5%) as discriminant taxa. Leuctridae (12.3%) reveal sites characterized by a lower amount of bank modification (CB_NAT) and no indicator taxa were detected here. At this node, BMS higher than the discriminant value (i.e. >18.5) hypothetically corresponds to a maximum of 60% of bank resectioning or 30% of bank protection. In practice, the threshold corresponds to $\approx 30\%$ bank altered. Group CB_NAT resulted as the most habitat-diversified and includes most of the natural or near-natural reference sites.

3.2.2. Environmental variables differing among MRT groups

A set of test variables was considered for better interpretation of MRT groups. For this purpose, BDM tests with post-hoc comparison were performed checking differences among MRT groups (results are presented in Table 2 and Table S3, SM). Spearman rank order correlation between variables is presented in Table S7, SM. For a thorough characterization of groups, minimum, median and maximum values of considered environmental variables are reported in Table S4, SM. Tree continuity diverged among MRT groups with the highest median values in CB_NAT and *cb_nat* groups. As far as habitat variables are concerned, significant differences among groups involved alders, channel shading and features related to channel substrate such as substrate diversity, with the highest values observed in CB_NAT and BA_MOD groups. Also, the percentage of organic channel substrates, in particular CPOM/Xylal, showed highly significant differences among groups; lowest median values were observed for groups *cb_mod*, *ch_mod* and *cb_nat*, all of them significantly different from groups CB_NAT and BA_MOD. Nearly significant differences ($p = 0.097$, Table 2; before FDR adjustment: $p = 0.007$) among MRT groups were also found for the combined presence of living parts of terrestrial plants (underwater fine roots), submerged macrophytes and mosses, almost absent in

altered sites. The same can be said for some mineral substrates ($p = 0.102$, Table 2; before FDR adjustment: $p = 0.008$): the most altered group (*cb_mod*) was characterized by a scarce presence of pebbles and cobbles. On the other side, the habitat-rich groups (CB_NAT and BA_MOD) were characterized by a high amount of sand (Table S4, SM). Cobbles were widely present in the *cb_nat* group whose sites are most subjected to important floods events (Table S4, SM). As far as flow characteristics are concerned, flow velocities were higher, and relatively turbulent flow types (Rippled and Unbroken Waves) more represented, in severely perturbed sites (*cb_mod*) than in habitat-rich and near-natural sites (Table S4, SM).

All CARAVAGGIO habitat descriptors but LRD, i.e. HMS, HQA, LUir and IQH, showed significant differences among groups. As far as individual land-use variables are concerned, significant differences among groups were highlighted for trees on both the bankface and over the banktop and for sparse vegetation, in adjacent lands only. Unsurprisingly, the reach-scale Land Use Index (LUir) showed high correlation with HMS and the presence of a range of artificial structures, usually established to defend uses. Differences among groups were not observed for larger scale variables, with notable exception of slope of the thalweg, for which group BA_MOD showed lowest median values. Also, no differences were revealed for water quality variables.

3.2.3. Benthic metrics response, differentiation among MRT groups and Ecological Status classification

The STAR_ICM index showed a positive correlation with tree cover continuity ($\rho = 0.61$) and with habitat descriptors HQA ($\rho = 0.53$) and IQH ($\rho = 0.54$) (Table S7, SM). Also, it is well correlated with channel and total reinforcement ($\rho = -0.50$ and -0.48). To a lesser extent, this macroinvertebrate index is also correlated to channel shading ($\rho = 0.43$) and organic and overall habitat diversity ($\rho = 0.38$ and 0.40 , respectively). The cumulative abundance of selected invertebrate taxa (i.e. Sel_EPTD) remarkably correlates with channel and bank modifications ($\rho \leq -0.50$ for a few variables).

All benthic metrics used in ecological status assessment showed legible differences among MRT groups (Table 3). For these metrics, the values observed in MRT groups were tested against reference conditions (sites isolated from groups *cb_nat* and CB_NAT). Significant results were found for STAR_ICMi, EPT, Sel_EPTD, Shannon and 1-GOLD. More in detail, Sel_EPTD is the metric that best discriminates, with all the three MRT groups containing altered sites being detected and differentiated from the expectation of unimpaired conditions. The STAR_ICM index clearly detected the most degraded conditions, in particular when both channel and banks modifications are present (*cb_mod*, $p < 0.05$), and when the main problem is channel modification and tree-related habitats are poor (*ch_mod*, $p < 0.1$). In somewhat similar conditions of morphological alteration but with well-developed tree-related habitats (BA_MOD), STAR_ICMi was unsuccessful in discriminating from unimpaired sites.

As far as Ecological Status classification (sensu WFD) is concerned, STAR_ICMi settled $\geq 60\%$ morphologically impaired sites of the *cb_mod* and *ch_mod* groups into the moderate status class (i.e. below the good status). For the BA_MOD group (abundant tree-related habitats) only one-third of sites showed a less than good benthic quality. Only one site belonging to the morphologically less impaired/unimpaired groups (*cb_nat* and CB_NAT) was classified below the good status.

3.2.4. Endemic species and MRT groups

In the detail of the single taxa (Table 4), four of them, i.e. *B. cf. albinatii*, *B. cf. ingradae*, *E. zebrata* and *A. cyrensis*, were found indifferently in all MRT groups showing no significant differences. *B. cyrneus* did not show any significant difference as well, although specimens were never collected in group '*cb_mod*'. The presence of *H. consiglioi* was different among groups; the species showed higher association with '*cb_nat*' group and was always absent from groups '*cb_mod*' and

'BA_MOD'. *I. insularis* was slightly associated to 'cb_nat' group and absent from 'cb_mod' group. Lastly, for *T. zavattarii*, an association to 'BA_MOD' group and absence from 'cb_mod' and 'ch_mod' groups were recorded.

4. Discussion

4.1. River modification, habitat features and MRT groups

Environmental variables used in MRT analysis embrace scattered bank and channel modification (resectioning and armouring) and crossing structures (bridges and culverts), wide-reaching in the Mediterranean area, together with natural features related to riparian vegetation and bank erosion. Only a few studies have directly examined possible negative impacts of channel and bank armouring, and related channel adjustments, on stream biotic communities (e.g. Stein et al., 2013). On the other hand, concordance between tree-related riparian features and benthic invertebrates is well documented in literature (e.g. Mason et al., 1984). According to Friberg (2014), "surprisingly few studies have documented clear impacts of habitat degradation" on biota, and this can probably partly be explained from the fact that habitat degradation is almost always occurring in combination with other types of pressure. In such circumstances, the effects of water pollution or multiple stressors can radically reduce our ability to disentangle the sole effect of degraded hydromorphology (Friberg, op. cit.). Also, the scientific community is in agreement that hydromorphological changes might not perturb macroinvertebrate communities of rivers to the same extent as stressors relating to water quality (e.g. Friberg et al., 2009, 2011). In this study, to better appreciate the effects of morphological alteration - intrinsically linked with habitat degradation - only sites not knowingly affected by other stressors (e.g. pollution, toxic substances, important water abstraction) were incorporated. Therefore, potential impacts such as water pollution (see Cao et al., 1996) and hydrological alteration (see Poff and Zimmerman, 2010), known to usually explain larger portions of variance for invertebrate assemblages, have been consciously excluded from this study. In the context of the performed MRT constrained analysis, overall model explanation was quite high (R^2 21%) and comparable with other MRT assessments dealing with biological community in streams (e.g. Obolewski et al., 2011). Similar percentages of explanation are found when e.g. multivariate ordination analyses are applied to assess the importance of environmental variables in structuring benthic communities (e.g. Feld and Hering, 2007; Petkovska and Urbanič, 2015). In many cases, the percentage of explanation found ranges from about 10% to 30% (e.g. Johnson et al., 2007). Test and inspection results for water quality variables, none of which significantly were associated to the MRT groups defined here, confirmed that the selection criteria for river reaches prevented the potential influence of water pollution on invertebrate communities. The same findings were obtained for other environmental and type variables (e.g. water temperature, discharge, distance from source). The only exception is the association of MRT groups with channel slope, obviously related to land use and, therefore, openly connected with the purpose itself of channel works or to the need for flood defence structures. As far as disturbance due to potential flooding is concerned, we expect that transversality of dates/months among the MRT groups corroborates a minor impact of hypothetical flooding on the community compared to changes due to river morphology.

Finally, sampling approach per se might limit the possibility of highlighting effects of morphological alteration on biota. To reduce biological unpredictability due to natural factors (i.e. the range and variability across seasons of micro-habitats observed along a pool-riffle sequence are expected to be larger than those noted in just one of the two mesohabitats), the adopted sampling protocol centres on the identification and sampling of the pool mesohabitat only (Buffagni et al., 2004). For instance, a strictly proportional multihabitat sampling including both pools and riffles might not be adequate in areas where

seasonal variation is relevant - like Mediterranean Europe -, because the resulting biological variability may mask the effect of pressures (Dallas, 2007; Leitão et al., 2014).

4.2. Do benthic communities reflect bank and channel modification in Mediterranean temporary rivers?

Channel stabilization through armouring (i.e. channel reinforcement) has resulted as the variable determining the first split of benthic community assemblages, separating the most impacted sites from others. Such sites present a high degree of bank alteration, too. Overall channel modification, including channel resectioning or reinforcement, culverts, fords, weirs and sluices, discriminated the second cluster of most altered river reaches from the remaining ones. Therefore, undeniably, a change in channel features from unimpaired conditions is directly and/or indirectly related to an analogous shift in invertebrate community in the studied river systems. Lower taxonomic variety and total abundance were observed in benthic fauna assemblages of channelized sites as well as higher flow velocities and presence of fairly turbulent flow types. These results are in accordance with some authors (Horsák et al., 2009; Wyżga et al., 2014) stating that rip-rap or concrete armours, where present, can increase flow velocity leading to a decrease of in-channel habitat heterogeneity and to a loss of invertebrate richness. Apart from the obvious difference in concrete/artificial substrate type (SP_CC in Table S4, SM), our results showed that sites affected by channel and bank stabilization differ in the presence of cobbles and pebbles (nearly absent in altered sites) and sand, which characterize habitat-rich rivers. Our findings support the general statement that differences in substrate, here probably related to the altered channel and bank conditions, may lead to the presence of distinctive habitats in the un-channelized sites, that can cause differences in taxonomic composition. In fact, substrate type is regarded as a major influence for invertebrate density and diversity (Beisel et al., 1998, 2000; Brooks et al., 2005; Graça et al., 2004). Likewise, differences were obtained for a range of organic substrates and particularly for CPOM/Xylal and for diversity of organic and combined organic/mineral substrates. Actually, the lowest amount and significantly smaller diversity of such habitats were observed in the two MRT groups characterized by important channel modifications. This fits with results found by Lepori et al. (2005) who demonstrated a low capacity of channelized stretches to retain organic detritus. We can thus expect here a direct influence of channel modification on invertebrate assemblages, e.g. by a reduction of number and extent of suitable habitats. For substrate diversity, encapsulating these two latter aspects, the direct influence of channel modification is remarkably corroborated by the response of the same Shannon-Wiener index applied to benthic taxa, that uniquely differentiates sites with modified channel from unimpaired sites (Table 3), and not those sites with degraded banks.

Actually, even if more easily classified according to the degree of channel modification, the two most impacted groups share a relatively high level of bank modification, and the effects of morphological alteration affecting these two river zones can hardly be disjointed. In the present study, high percentages of bank protection were present at most altered sites only (i.e. heavily modified water bodies) (Fig. 1), that were nonetheless discriminated by channel reinforcement. In this MRT group, the effects of channel and bank armouring concur, evidently affecting the benthic communities. When in place, bank protections can cause loss in lateral connectivity and limit habitat availability, thus reducing persistence and controlling density of invertebrates, especially during flood events (Negishi et al., 2002). In more general terms, even localized channelization - habitually affecting both channel and banks - can lead to smaller niches and refuges availability, especially due to a reduction of riparian vegetation (Schmetterling et al., 2001). In the study area, alterations caused by bank protection, with the consequent reduction of riparian refuges for adults of various taxa, might have improved the colonization by dominant eurytopic taxa as Empididae

(Rivosecchi, 1984), illustrative for two of the impacted groups. The colonization of Naididae may have been favoured as well (Kathman and Brinkhurst, 1998). Moreover, in the present study, some of the taxa associated to reinforced channel can be related to a considerable occurrence of peculiar organic habitats associated to channel armouring. For instance, high density of Gastropoda may be linked to the presence of conditions (e.g. concrete porosity) that could favour periphyton growth (even if we do not have direct evidence on this), the main food source for this class of molluscs (Obolewski et al., 2011). Likewise, the lower shading of the wetted channel observed in sites characterized by severe channel and bank modification could have favoured periphyton growth, creating suitable niches for Physidae.

4.3. Which habitats are more relevant in shaping benthic communities? The role of trees and the influence of bank resectioning

Tree-related bank and channel habitats emerge as the second most important factor in controlling benthic community. In the absence of morphological alteration, the highest total and EPT number of benthic families (N_families and EPT) were unambiguously observed at sites rich in tree-related habitats. The enlarged amount of CPOM and woody debris found there improves biodiversity as a consequence of increased structural complexity of the habitat (González and Graça, 2005), that may boost colonization potential for many aquatic species. Habitat-rich sites were also characterized by a higher amount of submerged macrophytes, living parts of terrestrial plants (underwater fine roots) and mosses, all contributing to sustain a high benthic diversification (e.g. González-Trujill, 2016). Moreover, an improved root formation due to the presence of trees along banks helps shaping riffle-pool morphology and increasing the roughness factor of the channel (Mason et al., 1984), that may support habitat diversification. In general terms, habitat richness and quality influence animal species diversity (e.g. Tews et al., 2004) and can largely shape aquatic communities (Greenwood et al., 2012; Quinn, 2005). This is especially evident when the share of riparian canopy varies (McCormick and Harrison, 2011), with a strong impact on tree-related features, e.g. shading (Broadmeadow and Nisbet, 2004), subsidies of organic matter to the river (Rios-Touma et al., 2014; Sudduth and Meyer, 2006) and woody debris dynamics (Quinn et al., 1997; Schmetterer et al., 2001). However, it is noteworthy to observe how two Ephemeroptera (Leptophlebiidae and Siphonuridae) and one Odonata (Libellulidae) families resulted as indicator taxa for the unimpacted sites showing a scarce range of tree-related habitats. Their presence seems basically related to the constant occurrence along the bankface and on adjacent lands of at least occasional clumps of trees i.e. to a certain degree of tree continuity (Table 2). At any rate, this is helpful in providing refuges from predation, favourable microclimatic conditions for adults and protection from desiccation during the adult stage, also finalising advantageous conditions for aquatic larval stages (Briers and Gee, 2004).

The two MRT groups with well-developed tree-related bank and channel habitats are divided by bank modification, mainly in terms of bank resectioning (Table S4, SM). The extent of trees (due to the presence of broad-leaved or Mediterranean forests, to mention some) along the bankface and on adjacent lands (mirroring less diffuse sparse vegetation), together with overall habitat quality (i.e. HMS and IQH), well differentiate habitat-rich clusters from each other. We can suppose that bank resectioning and associated reduction of tree cover accompanied by an increase in sparse vegetation and grasslands may have exerted a direct effect on habitats of benthic taxa at disturbed sites, given the importance of trees in improving bank stabilization by roots reinforcements, avoiding erosion or riverbank collapse and maintaining riverside habitats for species (Hubble et al., 2010). Bank resectioning is often observed in lowland and small rivers to control flood and land drainage, particularly in association to agricultural land use (Hughes et al., 2010; Raven et al., 1998). It may be concluded there is much evidence of the effect on invertebrates of tree-related features, mediated

by the degree of channel and bank alteration. In fact, the two MRT groups that include the most impacted sites are separated from others by a number of habitat variables related to presence and extent of trees along banks or in adjacent lands. Shading of channel, occurrence of alders on the lower bank, presence and continuity of tree cover all support the disjunction of the two groups from the other groups (Table 2).

4.4. Do bank and channel modification affect ecological status for aquatic invertebrates?

Notably, four of the six component metrics of the STAR_ICMi significantly discriminated at least one MRT group from the reference sites, and all groups were discriminated by inspecting that suite of metrics. Therefore, in the Mediterranean rivers of Sardinia, the impact of morphological alteration is detected by metrics in use for the assessment of ecological status (ES). Nonetheless, whereas statistical testing for STAR_ICMi was able to differentiate the two more altered groups, the null hypothesis could not be rejected for sites of cluster BA_MOD, where the presence of well-developed tree-related habitats presumably mitigated the effect of morphological alteration on benthic assemblages. In this regard, positive associations between macroinvertebrate metrics and forest land cover were found by Miserendino et al. (2011). It is worth noting that, in addition to the multimetric index, the three metrics showing significant separation from reference conditions take into account invertebrate abundance. 1-GOLD showed the lowest values in the three groups of macroinvertebrates associated to modified channel and/or banks. Low values of 1-GOLD, originally associated to organic pollution (Pinto et al., 2004), can reveal habitat alteration (Buffagni et al., 2007), as confirmed by the present study. Not surprisingly, the Shannon-Wiener index showed lower values in groups without tree-related habitats. In this regard, positive association between benthic taxon richness and diversity and increased tree cover in the riparian zone has been found (e.g. Rios and Bailey, 2006). Both 1-GOLD and the Shannon-Wiener index were already found to be significantly influenced by alteration of river morphology and habitat e.g. in terms of HMS and HQA, in Mediterranean rivers (Erba et al., 2015). The Sel_EPTD metric, expressing the sum of selected benthic taxa and designed to detect habitat modification (Buffagni et al., 2004, 2007), confirms its attitude - in terms of morphological alteration of both channel and banks. This abundance-based metric seems to have a large potential in detecting the effects of river modification on benthic taxa. A rise in macroinvertebrate abundance has been put in relation with reduced channel shading, due to increased light availability and primary production (Snyder and Johnson, 2006), that may be the case for river reaches of group cb_nat. Likewise, occurrence of in-channel organic habitats (Sudduth and Meyer, 2006) can explain the high values of this metric observed at sites in group CB_NAT. Interestingly and satisfactorily, differences in ASPT (Armitage et al., 1983), a metric mainly focused on water pollution, were not observed (Table 3). Neither did Total number of Families vary significantly, conceivably for its attitude to predominantly respond to larger scale factors (e.g. Erba et al., 2015).

As far as the classification of ES for the WFD is concerned, the STAR_ICM index ascribed to the moderate class (i.e. below the good ES) the large majority of sites that show evident morphological alteration. On the contrary, the classification based on benthic community did not entirely reflect the degree of morphological modification in less altered conditions, even if the STAR_ICM index actually showed a comparatively low median value, that corresponds to around one ES class difference. In general terms, even if the performance of the classification system for ES such as through the use of the STAR_ICMi, is considered acceptable, an upgrade can be envisaged. In fact, the most altered river stretches - where a very high degree of alteration is observed - do not fall into the possibly more appropriate poor and bad ES classes (they are all classified 'moderate'). Therefore, where morphological alteration is the main pressure, and other causes of degradation

with a potentially heavier impact (e.g. pollution and/or water abstraction) are not reported, the benthic response gradient used to classify ES might be stretched to better mirror the observed alteration range. What is more, a different weighing of metrics and/or the exclusion of bottom-performing ones can be considered.

4.5. Do endemic species mirror habitat features and river alteration?

The presence of a range of river modifications and habitat features characterizing MRT groups, distinctly influences some endemic species. In fact, *B. cyrneus*, *H. consilglio*, *I. insularis* and *T. zavattarii* were never collected from river reaches belonging to one or both the *cb_mod* and *ch_mod* groups, that include manifestly modified water bodies. Of these species, *B. cyrneus* and *H. consilglio* were absent from the third group of modified sites (*BA_MOD*) as well. Hence, the link to morphological alteration seems obvious, with the species *H. consilglio* and *I. insularis* associated to the *cb_nat* group. These results denote the studied types of bank and channel modification, including armouring and resectioning, as a potential threat for their persistence and conservation. Apart from its absence from the most altered group, *B. cyrneus* did not show any association to MRT groups. This may suggest a low sensitivity to slight bank and channel modification. *I. insularis* showed an association to the group characterized by unmodified sites lacking tree-related habitats. *B. cf. ingridae*, *E. zebrata* and *A. cyrnensis* were found indifferently in all groups, showing a non-significant response to the considered alterations and habitat features. This was somewhat predictable for *B. cf. ingridae*, because this taxon is likely vicariant for the eurytopic species *Baetis rhodani* (Pictet, 1843) in Corsica (Thomas and Soldán, 1987) and Sardinia. Also, more than one species of the *rhodani* group have been recently hypothesized based on genetic information (Bisconti et al., Unpublished data) and species attribution based on morphological characters will have to be re-assessed. To highlight the association between the endemic taxa - and their potential regression - and MRT groups (i.e. different levels of morphological alteration and tree-related habitats), sampling covered all major habitats present in the reach (e.g. Harding, 2003; Strayer, 2006) along a pool-riffle sequence and included river margins, submerged roots and woody debris.

Remarkably, at least in areas like Sardinia with high values of endemism richness and where species conservation status is largely unknown, biodiversity conservation should involve more species than only those listed in the HABITATS Directive. On this subject, the Mediterranean Basin which is one of the 25 biodiversity hotspots of the world, should be considered as a hyper-hot candidate for conservation support for its exceptional total of endemic taxa (e.g. Myers et al., 2000). Here, on a smaller scale, Tyrrhenian islands are one of the hotspots (Médail and Quézel, 1999). The exceptional concentration of endemic species will likely suffer large losses of habitat, also in relation to climate change (New, 2008). Actually, many island insects live only in remnant habitats, and protecting these is a key theme in planning insect conservation on islands (New, 2008). The results obtained in the present study directly match the presence of endemic species with the degree of morphological alteration of rivers and their ecological status. Any in-depth investigation of habitat factors influencing each species is beyond the scopes of this paper, but the unique rate of endemism in Sardinian rivers and the paucity of existing information, when compared to other Mediterranean-climate areas (e.g. Ball et al., 2013), call for further research to support sound biodiversity actions and species conservation.

5. Implications for management and conclusion

5.1. Can we improve the effectiveness of stream corridor management practices for aquatic invertebrates?

In this paper, the relevance of river attributes that can be directly controlled by water managers within management plans was

evaluated. Thresholds for the presence of artificial intervention of various kinds and the extent of natural features along the river channel were recognized and offered, which can be used to guide scheming and the implementation of related measures. These are: the extent of i) channel reinforcement (if $\geq 40\%$, the community is critically depleted), ii) channel modification (10%) and iii) bank modification (a BMS score ≈ 18.5 is a critical limit). Moreover, the association among tree-related habitat features, which can mitigate the negative effect of channel and bank modification when their average is $\geq 5\%$, and benthic taxa has been demonstrated and emphasized.

Interestingly, the extent of tree-related habitats differentiates two pairs of groups, both including one cluster of nearly-natural reaches and one of altered sites. After most degraded sites are removed in the studied river systems, the influence of tree-related habitats on aquatic invertebrates is then stronger than that of morphological alteration per se. This looks very promising, because a variety of decision-making options, that could result appropriate to protect benthic communities, may emerge from the analysis of the possible measures for streams that need to be restored. For instance, the presence of weirs and road crossings, rather ubiquitous and predominant in Mediterranean, rural environments (e.g. RAS, 2005), determines changes to river channels (Gregory, 2006) and in-channel habitats disposal and arrangement (e.g. Litvan et al., 2008; Mueller et al., 2011). Even if less intrusive approaches for new erections can be adopted, for most of the installations already present, removal or important ameliorations to their physical structure are hardly foreseeable. The same can be said for bank armouring. As a consequence, establishment and sustainable management of vegetation in the riparian zone of rivers can be seen as an opportune, achievable prospect. In this paper, a number of habitat factors emerged, whose presence should be promoted in the rivers of Sardinia to support both Ecological Status and the possible occurrence of endemic species. These are (first most relevant): 1) presence of trees in the river corridor, 2) presence of trees in adjacent lands, 3) continuity of tree cover in the river corridor, i.e. at least occasional clumps of trees (≥ 2 m high) should be guaranteed. Additionally, in river systems where instability of stream reaches is not too high (see Section 5.2), shading of river channel (i.e. $\geq 40\%$) and the presence of alders should be maintained, paralleled by the goal of retaining/increasing diversity of in-stream organic habitats, especially CPOM, Xylal, living parts of terrestrial plants (TP) and submerged macrophytes. The importance of riparian vegetation in providing multiple ecosystem services is actually recognized worldwide (Millennium Ecosystem Assessment, 2005). In this context, the role of tree plantations is crucial in restoring habitat for biodiversity (Astudillo et al., 2016; Nimmo et al., 2016; Surasinghe and Baldwin, 2015), as well as to buffer the influx of agricultural pollutants entering streams (Balestrini et al., 2016; Sabater et al., 2003).

5.2. Distinctiveness of Mediterranean river systems and setting priorities for aquatic invertebrates

In this study, we produced plain evidence that the reach scale matters in Mediterranean temporary streams. Nonetheless, larger spatial - and temporal - scales should also be considered in setting management priorities. In fact, in the semiarid Mediterranean climate, recovery of channel form tends to be in the order of decades and extreme events with long recurrence intervals (e.g. 50 years) may cause important and long-lasting geomorphic changes (De Waele et al., 2010). Explicitly in a Karstic environment, overbank flow can determine significant changes in channel morphology and scour the vegetation on the valley bottom, as it was observed at the Flumineddu site (MRT group *cb_nat*), as a consequence of the 2004 extraordinary flood (De Waele et al., 2010). This river can exemplify the situation of many of the sites in the same group, which includes six reference sites, where tree-related habitats are nearly absent, possibly for natural reasons such as meteorological events and/or climatic constraints. Here, biological elements may reach lower richness and diversity (Table 3) than in sites less affected by

natural disturbance. In general terms, this may be at least partly related to instability of stream reaches, that may provide worse physical habitat and lower levels of in-stream habitat structural diversity and quality than stable ones (Mažeica et al., 2004). Armouring of streambanks, to prevent infrastructure from flooding or erosion, leads to increased local bank stability. However, bank erosion is a desirable attribute for rivers, because it is a geomorphic process that upholds successions of riparian vegetation and maintains dynamic habitats, integral to the functioning of the ecosystem (Florsheim et al., 2008). Therefore, even if individual biological indicators (e.g. richness) may get a higher score, stable reaches should not be considered a management target per se and the general links between geomorphic processes, including adjustment ones, and ecological functions should be established to support sustainable river and watershed management (Florsheim et al., 2008). There are various uncertain aspects to be taken into account when assessing the best options. Tree-related habitats can be more easily maintained or recreated in stable channels. On the contrary, in rivers where high levels of disturbance contribute to deeply and/or frequently alter tree-related components and habitats, management objectives and measures should be deliberated with specific attentiveness.

The natural ecosystem services, including the need to preserve endemic species, should be similarly assessed on a larger scale, for example at the valley or catchment level. In fact, freshwater conservation should better manage entire watersheds and move towards broader approaches that try to concurrently safeguard human needs for fresh water and as much biodiversity as possible (Strayer, 2006). Even if the importance of endemic species for biodiversity is well recognized, their inclusion in WFD management plans is not always considered. The introduction of management criteria that openly use information on endemic species seems therefore crucial to reduce the risk of destroying biodiversity in the attempt to improve overall ecological status. For instance, in Sardinia, an incorrect management of river systems where habitats are naturally poor, can increase the threat towards the endemic species *Habrophlebia consilgloi* and *Isoptera insularis*. This is due to the fact that these are mostly found in unimpaired and unpolluted river reaches where tree-related bank and channel habitats are poorly represented.

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.scitotenv.2016.05.154>.

Acknowledgements

This study was carried out within the LIFE + INHABIT project (LIFE08 ENV/IT/000413): 'Local hydro-morphology, habitat and RBMPs: new measures to improve ecological quality in South European rivers and lakes', under the LIFE Programme of the European Union. Part of the data was collected within the MICARI project (MIUR, D.M. 408 Ric. 20.03.2002 – Settore "Risorse Idriche"). We thank all the colleagues from Regione Autonoma della Sardegna (RAS), and particularly Dr. Rita Casula, who was intensively dedicated in giving her assistance during the organization of the study. We are grateful to the anonymous reviewers for useful comments on an earlier version of the manuscript.

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